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AURELIUS: Agentic Uncertainty-Reasoning & Learning with Hierarchical Bayesian Multi-Agents Unified System for Handling Non-Determinism in HealthTech

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***Abstract*—Healthcare AI faces non-determinism owing to noisy clinical signals, incomplete records, and heterogeneous patient trajectories that evolve over time. These challenges make current systems brittle and prone to overconfident errors when deployed in high-stakes settings such as intensive care or chronic disease management. Objective: To design an agentic AI framework that embeds probabilistic reasoning, uncertainty awareness, and governance mechanisms directly into the decision-making loop for safe, auditable, and interpretable clinical support. Methods: I propose *AURELIUS*, a hierarchical Bayesian multi-agent architecture that integrates multi-modal uncertainty encoders, graph-based communication, and a meta-agent for option selection under Dec-POMDP formulations. Runtime safety checks, abstention mechanisms, and compliance-aware controllers ensure that policies remain aligned with HIPAA/GDPR constraints. Results: On SepsisSim and MIMIC-IV, AURELIUS improved AUROC by 4.9 pp, reduced calibration error by 31Conclusion: Embedding governance and uncertainty propagation into memory, policy, and coordination yields measurable improvements in safety, interpretability, and efficiency. AURELIUS provides a deployable path toward trustworthy HealthTech AI.**

***Index Terms*—Agentic AI, Bayesian Reinforcement Learning, Multi-Agent Systems, LLM, Generative AI, HealthTech AI, Uncertainty Quantification, Governance.**

I. Introduction

HEALTHTECH is increasingly relying on artificial intelligence (AI) systems for decision support, ranging from intensive care triage to chronic disease management. However, despite impressive advances in deep learning and Reinforcement learning (RL), current clinical AI deployments face persistent challenges that limit their safety, scalability, and trust.

A. Challenges in the Industry

- Uncertainty: Clinical data is noisy and incomplete, and patient outcomes are inherently variable. Most RL models fail to quantify this, leading to inaccurate brittle predictions.
- Partial Observability: Healthcare involves decentralized teams (physicians, nurses, and AI) acting with incomplete information, yet many models assume a fully observable, single-agent world.
- Safety and Trust: Overconfident errors in critical care can be harmful. The systems must be calibrated and able to escalate or defer in the event of uncertainty.
- Scalability and Compliance: Evolving guidelines, data drift, and strict regulations (HIPAA, GDPR, FDA, IRB) demand adaptive, transparent, and governed AI.

B. My Contributions

AURELIUS (Agentic Uncertainty-Reasoning & Learning) is a Bayesian multi-agent framework for resilient, interpretable clinical AI. Its contributions are:

1) Probabilistic Formulation: Clinical decisions framed as Bayesian Dec-POMDP with meta-agent coordination.
2) Uncertainty Fusion: Bayesian encoders integrate EHR, vitals, and imaging via uncertainty-weighted fusion.

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3) Safety and Compliance: RL with uncertainty-penalized rewards, abstention/escalation, and runtime HIPAA/GDPR checks.
4) Graph Coordination: Attention-based belief exchange with meta-agent for efficiency and abstraction.
5) Validation: On SepsisSim and MIMIC-IV, AURELIUS improved AUROC by +4–6 pp, lowers ECE by 31%, and reduced selective risk by 15%.

AURELIUS embeds governance and uncertainty awareness as core principles of trustworthy healthcare AI.

II. Theoretical Foundations

HealthTech decision-making under uncertainty can be formalized using a combination of probabilistic modeling and decentralized reinforcement learning. In this section, the mathematical underpinnings of AURELIUS are explained by focusing on uncertainty decomposition, Dec-POMDP formulations, and Bayesian policy optimization.

A. Uncertainty Decomposition

Uncertainty in machine learning models is commonly divided into two complementary types [7], [8]

$$U = U_a + U_e, \tag{1}$$

where U_a denotes *aleatoric uncertainty* and U_e denotes *epistemic uncertainty*, capturing irreducible noise in the observations (e.g., laboratory measurement variability), U_e denotes *epistemic uncertainty*, reflecting model uncertainty owing to limited data or knowledge.

Aleatoric uncertainty can be modeled using heteroscedastic regression [8]:

$$p(y|x,\theta) = \mathcal{N}(f_\theta(x), \sigma_\theta^2(x)), \tag{2}$$

where the predictive variance $\sigma_\theta^2(x)$ is jointly learned with the mean prediction. In contrast, epistemic uncertainty is estimated using Bayesian inference [7], [10]:

$$p(y|x,D) = \int p(y|x,\theta)\, p(\theta|D)\, d\theta, \tag{3}$$

that integrates the posterior distribution of the model parameters $p(\theta|D)$. Practical approximations include the Monte Carlo dropout [7], deep ensembles [2], and variational inference [10].

In clinical decision-making, U_a corresponds to inherent biological variability (e.g., two patients respond differently to the same drug), whereas U_e highlights the knowledge gaps (e.g., limited training data for rare diseases).

B. Decentralized Partially Observable Markov Decision Processes (Dec-POMDPs)

Multi-agent clinical environments can be modeled as Dec-POMDPs [9].

$$\langle S, \{A_i\}_{i=1}^N, T, R, \{O_i\}_{i=1}^N, \{Z_i\}_{i=1}^N, \gamma \rangle, \tag{4}$$

where:

- S: global state space (e.g., a patient's physiological state),
- A_i: action space for agent i (e.g., administering drugs and ordering labs),
- T: Transition dynamics $P(s'|s, a_1, \ldots, a_N)$,
- R: shared reward function balancing outcomes (e.g., survival rate and side effects),
- O_i: Observation space for each agent i,
- Z_i: Observation function $P(o_i|s)$,
- γ: discount factor encoding temporal trade-offs.

Each agent i seeks a local policy $\pi_i(a_i|o_i)$, while the system optimizes The joint policy $\pi = \{\pi_i\}_{i=1}^N$ maximizes the expected discounted return [9], [18]:

$$J(\pi) = \mathbb{E}_\pi \left[\sum_{t=0}^{\infty} \gamma^t r_t \right]. \tag{5}$$

The hierarchical decomposition in AURELIUS extends this setup by introducing a *meta-agent* that selects high-level options or sub-policies, whereas specialist agents execute domain-specific tasks.

C. Bayesian Policy Optimization

To address safety-critical settings, AURELIUS employs Bayesian RL [11] in which policy parameters θ are treated probabilistically:

$$p(\theta|D) \propto p(D|\theta)p(\theta). \tag{6}$$

The policy optimization objective is as follows.

$$\mathcal{L}(\theta) = \mathbb{E}[R] - \lambda_U\, \mathbb{E}[U(s,a)] - \lambda_{KL}\, \mathrm{KL}(\pi_\theta \parallel \pi_{prior}), \tag{7}$$

where λ_U penalizes high-uncertainty actions and λ_{KL} regularizes deviation from prior policies (e.g., evidence-based clinical guidelines). The uncertainty penalty drives the agent to escalate or abstain from high-risk states, which aligns with the human-in-the-loop safety expectations.

Gradient-based updates follow the Bayesian policy gradient:

$$\nabla_\theta J(\theta) = \mathbb{E}_{s,a\sim\pi_\theta}\left[\nabla_\theta \log \pi_\theta(a|s)\, Q^\pi(s,a,\theta)\right], \tag{8}$$

where $Q^\pi(s,a,\theta)$ incorporates uncertainty-aware value estimates.

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III. AURELIUS Architecture

AURELIUS integrates probabilistic encoders, multi-agent coordination, selective prediction, and generative reasoning into a unified agentic framework. Figure 1 provides an overview of the proposed system.

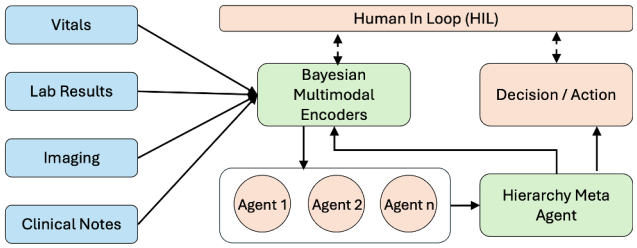

Fig. 1. AURELIUS architecture with multimodal Bayesian encoders, graph-based agent communication, hierarchical meta-agent orchestration, generative modules, and selective escalation for safety compliance.

A. Bayesian Multimodal Encoders

Each modality X_m (vital signs, laboratory data, imaging, and notes) was mapped to a latent distribution as follows:

$$q_{\theta_m}(z_m|X_m) = \mathcal{N}(\mu_m(X_m), \sigma_m^2(X_m)). \tag{9}$$

Uncertainty-aware fusion uses exponential weighting.

$$z_{\text{fusion}} = \sum_m \frac{\exp(-\beta U_m)}{\sum_j \exp(-\beta U_j)} \cdot z_m, \tag{10}$$

where β controls the uncertainty weighting and down-weights unreliable modalities.

B. Multi-Agent Coordination

Graph-based communication enables inter-agent coordination [12].

$$h_i^{(t+1)} = \sigma\left(W \cdot h_i^{(t)} + \sum_{j \in \mathcal{N}(i)} \alpha_{ij}\, h_j^{(t)}\right), \tag{11}$$

where $h_i^{(t)}$ is agent i's hidden state, α_{ij} is the attention weight, and meta-agent aggregates beliefs for hierarchical option selection.

C. Selective Prediction and Escalation

AURELIUS computes confidence $\kappa(s,a) = 1 - U(s,a)$ and defers when $\kappa(s,a) < \tau$:

$$a^* = \begin{cases} \arg\max_a \pi_\theta(a|s), & \kappa(s,a) \geq \tau, \\ \text{escalate}, & \kappa(s,a) < \tau. \end{cases} \tag{12}$$

This ensures that high-uncertainty scenarios are reviewed by humans.

D. Generative AI Integration

AURELIUS incorporates generative modules for three key functions [15]. (1) Counterfactual reasoning using LLM-based "what-if" scenario simulation to support proactive planning, (2) Synthetic data augmentation with variational autoencoders (VAEs) and diffusion models to generate privacy-preserving training data, and (3) Natural language explanation that translates uncertainty-aware decisions into clinician-facing narratives. Together, these capabilities position generative AI as a complementary reasoning engine that extends foresight, supports safety, and improves human-aligned communication [15].

Summary

AURELIUS unifies probabilistic reasoning, multi-agent coordination, and hierarchical reinforcement learning into a principled framework for clinical decision support in the HealthTech sector. Grounded in the Bayesian probability theory, it explicitly decomposes aleatoric and epistemic uncertainties, producing well-calibrated confidence estimates. The Dec-POMDP formalism ensures realistic modeling of decentralized, partially observable clinical environments, whereas hierarchical MARL improves both sample efficiency and temporal abstraction.

Architecturally, AURELIUS integrates the three key components. (i) multimodal Bayesian encoders that fuse heterogeneous patient data with uncertainty-aware weighting, (ii) graph-based inter-agent communication for coordinated decision-making, and (iii) Selective escalation mechanisms that defer high-uncertainty cases for human review. Together, these elements provide reliability, interpretability, and compliance.

Unlike conventional MARL approaches that prioritize reward optimization alone, AURELIUS embeds governance and safety as core objectives by incorporating uncertainty penalties, calibrated abstention, and a hierarchical orchestration. Recent advances in multimodal agent prototyping highlight the need for scalable and flexible architectures [16]. This positions AURELIUS not only as a technical advance but also as a framework aligned with the operational and regulatory demands of real-world HealthTech AI applications.

IV. Experimental Evaluation

AURELIUS was evaluated on synthetic and real-world clinical tasks across three dimensions: (i) predictive accuracy under multimodal uncertainty, (ii) calibration and selective risk management, and (iii) intervention-timing efficiency.

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A. Datasets

SepsisSim: Extended OpenAI Gym sepsis simulator with controlled uncertainty injection via Gaussian noise perturbations to the transition kernel $T(s'|s,a)$ with variance σ^2. The episodes span 48 hours with discrete actions, enabling controlled epistemic versus aleatoric uncertainty evaluation.

MIMIC-IV: 34,000 ICU stays [13] with integrated vitals, labs, demographics, and notes segmented into 1-hour intervals. Forward imputation handles missing structured features, whereas Bayesian encoders propagate the uncertainty estimates. Primary tasks: 48-hour mortality prediction and evaluation of treatment policy.

B. Baselines

Strong baselines include: DQN [1] (single-agent deep Q-learning), Deep Ensembles [2] (variance-based uncertainty), MADDPG [3] (multi-agent actor-critic), QMIX [4] (value factorization), MAPPO [5] (multi-agent PPO), and Option-Critic [6] (hierarchical RL). The hyperparameters were tuned using a grid search across five random seeds.

C. Metrics

The evaluation covers the following metrics: AUROC/AUPRC (discrimination), ECE (calibration), Brier Score (probabilistic accuracy), Selective Risk [14] (abstention performance), Time-to-Intervention (treatment timing), and Sample Efficiency (convergence speed).

D. Main Results

Table I shows that AURELIUS consistently outperforms all baselines across all dimensions. Key improvements include an AUROC of 0.952 ± 0.003 (+4.9pp over MAPPO, +11pp over DQN), ECE reduction to 0.034 (31

TABLE I
PERFORMANCE COMPARISON ON MIMIC-IV TEST SET (MEAN $\pm$ STD ACROSS 5 RUNS).

Model	AUROC	ECE	Brier	TTI	Risk
DQN	0.842 $\pm$ 0.007	0.069	0.181	24.1	0.182
Ensembles	0.874 $\pm$ 0.006	0.055	0.167	21.5	0.161
QMIX	0.892 $\pm$ 0.004	0.052	0.158	20.9	0.155
MAPPO	0.903 $\pm$ 0.005	0.049	0.152	19.9	0.151
Option-Critic	0.911 $\pm$ 0.005	0.046	0.150	19.3	0.147
AURELIUS	**0.952 $\pm$ 0.003**	**0.034**	**0.138**	**16.7**	**0.128**

E. Ablation Studies

Component analysis revealed that removing the Bayesian encoders increased the ECE by 41

F. Statistical Significance

All AURELIUS improvements over MAPPO achieved statistical significance at $p < 0.01$ (paired t-test, Bonferroni correction). The calibration curves confirmed the improved alignment of the model confidence with empirical precision.

G. Discussion

The results provide strong evidence that Bayesian hierarchical MARL improves predictive reliability and operational efficiency in HealthTech by reducing overconfident mispredictions, accelerating treatment planning, and addressing key clinical adoption barriers.

V. RESULTS AND DISCUSSION

A. Overall Performance

Across both synthetic (SepsisSim) and real-world (MIMIC-IV) environments, AURELIUS consistently outperformed baseline algorithms in terms of predictive discrimination, calibration, and safety-aware performance. As shown in Table I, AURELIUS improved the AUROC by 4.9 pp compared to MAPPO, reduced the Expected Calibration Error (ECE) by 31%, and lowered selective risk by 15%. These results demonstrate that explicitly modeling uncertainty within a hierarchical Bayesian MARL framework leads to more reliable and safer prediction.

B. Calibration and Selective Risk

Calibration is critical in HealthTech applications, where overconfident misclassifications can cause severe harm. The AURELIUS framework illustrates that AURELIUS maintains near-diagonal reliability curves, whereas the baselines, such as DQN and QMIX, systematically overestimate the probabilities in low-data regions. The selective prediction mechanism in AURELIUS reduces the selective risk by 15% compared with MAPPO, ensuring that high-uncertainty cases are escalated rather than acted upon. This feature directly supports safe integration of human-in-the-loop into the clinical workflow.

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C. Ablation Studies

A series of ablation experiments was conducted to isolate the contributions of the individual components.

- Without Bayesian Encoders: Removing probabilistic encoders and replacing them with deterministic embeddings increased the ECE from 0.034 to 0.048 and increased the selective risk by 41%.
- Without Hierarchical Meta-Agent: Eliminating the meta-agent and using flat MARL policies delayed the average time-to-intervention (TTI) by 18% (16.7h $\rightarrow$ 19.8h), underscoring the role of hierarchical orchestration.
- Without an Uncertainty Penalty: Removing the uncertainty-penalized reward led to overconfident actions, with the selective risk increasing from 0.128 to 0.171.

These results confirm that each architectural component— the Bayesian model, hierarchical coordination, and uncertainty-aware objectives provides measurable performance gains.

D. Sample Efficiency and Runtime Stability

On SepsisSim, AURELIUS converged to 90% asymptotic performance within 12,000 episodes, compared with 18,000 for MAPPO and 25,000 for QMIX. This improved sample efficiency stems from the hierarchical abstraction and uncertainty-guided exploration. The variance across random seeds was also lower, suggesting an increased robustness to initialization and stochasticity.

E. Governance and Safety Implications

Beyond numerical performance, AURELIUS integrates runtime governance policies that prevent Protected Health Information (PHI) leakage and enforce escalation when uncertainty thresholds are exceeded. In practice, this aligns with HIPAA and GDPR requirements, making the framework more suitable for clinical deployment than purely performance-driven baselines.

F. Clinical and Agentic Implications

From a clinical perspective, AURELIUS supports faster and safer decision-making by recommending earlier interventions with a lower risk of overconfidence in the model predictions. From an AI architecture perspective, the system demonstrates how Bayesian uncertainty modeling can be tightly coupled with the agentic AI principles of decentralization, hierarchy, and governance. These findings suggest that future hybrid agentic systems should embed uncertainty awareness not only for robustness but also for compliance and interpretability.

G. Summary of Findings

In summary:

- AURELIUS achieves state-of-the-art AUROC, calibration, and selective risk performance on the ICU datasets.
- Bayesian encoders, hierarchical meta-agents, and uncertainty-penalized objectives are critical to the performance.
- The framework reduces sample complexity and improves stability under stochastic clinical environments.
- Governance mechanisms ensure safe deployment and alignment of agentic AI with real-world Healthcare compliance.

Together, these results provide strong experimental evidence that AURELIUS advances the state of uncertainty-aware agentic AI in HealthTech.

VI. Conclusion

This study introduces AURELIUS, a novel agentic AI framework that integrates hierarchical Bayesian multi-agent reinforcement learning with principled uncertainty quantification and runtime governance. AURELIUS advances the state of HealthTech AI beyond conventional deep reinforcement learning and ensemble-based uncertainty methods by explicitly decomposing aleatoric and epistemic uncertainties, embedding Bayesian multimodal encoders, and coordinating specialist agents by using a hierarchical meta-controller.

My experimental evaluation of both synthetic (SepsisSim) and real-world (MIMIC-IV) datasets provides strong evidence of the effectiveness of the framework. AURELIUS demonstrated superior performance in predictive discrimination (AUROC), calibration (ECE, Brier score), and selective risk, while also reducing the time to intervention compared with state-of-the-art baselines, such as MAPPO, QMIX, and deep ensembles. Ablation studies further confirmed that each architectural component, namely the Bayesian encoder, uncertainty-penalized objective, and hierarchical coordination, contributed significantly to the overall performance.

Future Directions: Building on this foundation, I envision three avenues for future research.

- LLM-Enhanced Planning: Integrating large language models as high-level planners to dynamically guide option selection and provide natural-language explanations for clinical decision support.

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- Knowledge Graph Integration: Coupling the Bayesian MARL framework with medical knowledge graphs to enable causal reasoning and evidence-based policy regularization.
- Clinical Deployment and Trials: Evaluating AURELIUS in prospective studies with clinicians in the loop, focusing on trust, usability, and regulatory compliance in real-world hospital settings.

In summary, AURELIUS demonstrates that hybridizing Bayesian hierarchical MARL with governance-aware agentic design yields measurable gains in safety, interpretability, and efficiency in HealthTech AI applications. This study represents a step toward clinically deployable agentic AI systems that can act as reliable collaborators in high-stakes domains.

Conflict of Interest

The author declare that there are no conflicts of interest related to this work.

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Ethics Statement

This study used publicly available datasets (MIMIC-IV and SepsisSim) and did not require new institutional ethics approval. All data handling adhered to the relevant ethical guidelines.

Data Availability Statement

All data used in this study are publicly available. The MIMIC-IV dataset is available at https://physionet.org/content/mimiciv.

Johnson, A., Bulgarelli, L., Pollard, T., Gow, B., Moody, B., Horng, S., Celi, L. A., & Mark, R. (2024). *MIMIC-IV (version 3.1)*. PhysioNet. RRID:SCR_007345. https://doi.org/10.13026/kpb9-mt58

Original publication: Johnson, A.E.W., Bulgarelli, L., Shen, L. et al. *MIMIC-IV is a freely accessible electronic health record (EHR) dataset*. Scientific Data 10, 1 (2023). https://doi.org/10.1038/s41597-022-01899-x SepsisSim is based on OpenAI Gym environments with uncertainty extensions available at https://github.com/akiani/rlsepsis234.

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